Estatística

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a) $\mu = \bar{x} \pm Z\frac{\alpha}{2}\left(\frac{\sigma}{\sqrt{n}}\right)$ $\mu = 210 \pm Z_{0,025} - \frac{13}{\sqrt{30}}$

$\rightsquigarrow \mu = 210 \pm 1,96 \cdot \frac{13}{\sqrt{30}}$ $\begin{cases} \mu = 205,348 \\ \mu = 214,652 \end{cases}$ $(205,348 ; 214,652)$

b) $\mu = 210 \pm 2,376 \cdot 13/\sqrt{30}$ $Z_{0,005} = 2,576$

$\mu = 203,886$

$\mu = 216,114$

$(203,886 ; 216,114)$

c) $E = 3$ $E = Z \cdot \frac{\sigma}{\sqrt{n}}$ $3 = 1,96 \cdot 13/\sqrt{n} \rightsquigarrow \sqrt{n} = \frac{1,96 \cdot 13}{3}$

$\rightsquigarrow n \left(\frac{1,96 \cdot 13}{3}\right)^2 = 72,137$

$\rightsquigarrow 72,137 - 30 = [42,137] \rightsquigarrow 43$ copos